

Home Labe Week 5, Day 2 revised

Task:

Execute a simulated FTP or HTTP login from Kali Linux to Ubuntu while capturing the network traffic with Wireshark.

Simulated FTP

```
nmap -sV -p 21 192.168.78.5
```

We can clearly see that port 21 is open.

21/tcp open ftp vsftpd 3.0.5

```
(kali㉿kali)-[~]
└─$ nmap -sV -p 21 192.168.78.5
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-20 21:52 -0400
Nmap scan report for 192.168.78.5
Host is up (0.0014s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
MAC Address: 08:00:27:E3:80:BC (Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.15 seconds
```

Now let's create a dedicated lab ftp account on ubuntu

Username: ftpser

Password: msfadmin

```

vboxuser@Ubuntu:~$ sudo adduser ftpuser
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
^[[Asudo: maximum 3 incorrect authentication attempts
vboxuser@Ubuntu:~$ sudo adduser ftpuser
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
esudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: maximum 3 incorrect authentication attempts
vboxuser@Ubuntu:~$
vboxuser@Ubuntu:~$ sudo adduser ftpuser
[sudo: authenticate] Password:
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for ftpuser
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
vboxuser@Ubuntu:~$

```

Now in Kali terminal type

ftp 192.168.78.5

When prompted enter username

username is: ftpuser

For password prompt enter : msfadmin

Note: On **Kali**, we need to establish FTP connection using the same **Ubuntu's username we created to log into Ubuntu**:

```

(kali㉿kali)-[~]
└─$ ftp 192.168.78.5
Connected to 192.168.78.5.
220 (vsFTPd 3.0.5)
Name (192.168.78.5:kali): ftpuser
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>

```

We were told that login was successful

At that point the task requirement is demonstrated:

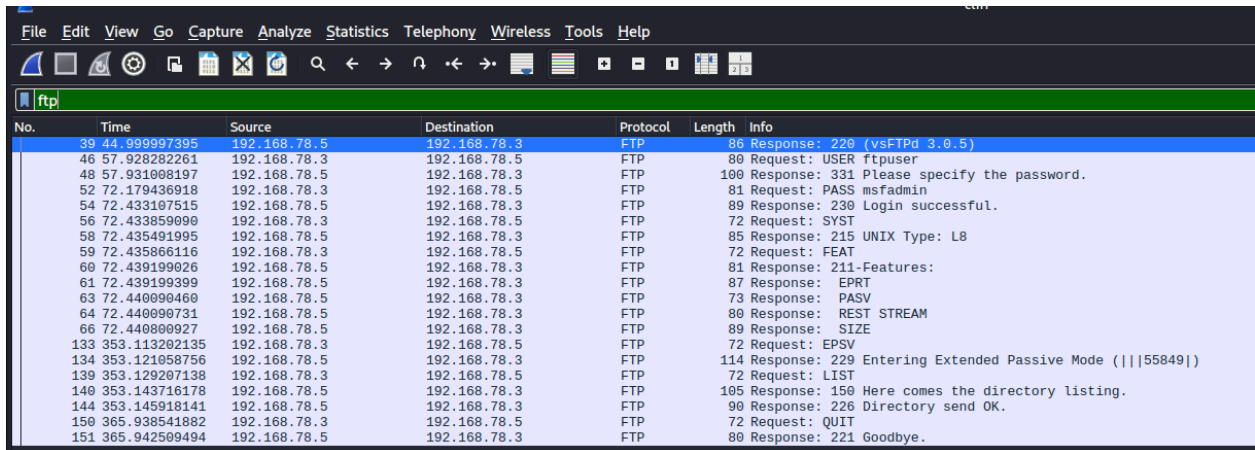
Kali → Ubuntu → FTP → TCP/21

To verify the FTP session we type the command: ls

Then we type: quit to exit.

Now in wireshark we can generate the FTP traffic and capture it for **analysis**.

To filter ftp traffic we type ftp.



No.	Time	Source	Destination	Protocol	Length	Info
39	44.999997395	192.168.78.5	192.168.78.3	FTP	86	Response: 220 (vsFTPd 3.0.5)
46	57.928282261	192.168.78.3	192.168.78.5	FTP	80	Request: USER ftpuser
48	57.931008197	192.168.78.5	192.168.78.3	FTP	100	Response: 331 Please specify the password.
52	72.179436918	192.168.78.3	192.168.78.5	FTP	81	Request: PASS msfadmin
54	72.433107515	192.168.78.5	192.168.78.3	FTP	89	Response: 230 Login successful.
56	72.433859090	192.168.78.3	192.168.78.5	FTP	72	Request: SYST
58	72.435401995	192.168.78.5	192.168.78.3	FTP	85	Response: 215 UNIX Type: L8
59	72.435866116	192.168.78.3	192.168.78.5	FTP	72	Request: FEAT
60	72.439199026	192.168.78.5	192.168.78.3	FTP	81	Response: 211-Features:
61	72.439199399	192.168.78.5	192.168.78.3	FTP	87	Response: EPRT
63	72.440090460	192.168.78.5	192.168.78.3	FTP	73	Response: PASV
64	72.440090731	192.168.78.5	192.168.78.3	FTP	80	Response: REST STREAM
66	72.440809027	192.168.78.5	192.168.78.3	FTP	89	Response: SIZE
133	353.113202135	192.168.78.3	192.168.78.5	FTP	72	Request: EPSV
134	353.121058756	192.168.78.5	192.168.78.3	FTP	114	Response: 229 Entering Extended Passive Mode (55849)
139	353.129207138	192.168.78.3	192.168.78.5	FTP	72	Request: LIST
140	353.143716178	192.168.78.5	192.168.78.3	FTP	105	Response: 150 Here comes the directory listing.
144	353.145918141	192.168.78.5	192.168.78.3	FTP	90	Response: 226 Directory send OK.
150	365.938541882	192.168.78.3	192.168.78.5	FTP	72	Request: QUIT
151	365.942509494	192.168.78.5	192.168.78.3	FTP	80	Response: 221 Goodbye.

The password is in plain text

Length	Info
86	Response: 220 (vsFTPd 3.0.5)
80	Request: USER ftpuser
100	Response: 331 Please specify the password.
81	Request: PASS msfadmin
89	Response: 230 Login successful.
72	Request: SYST

Simulated HTTP

Stage 1 — Set up the HTTP server on Ubuntu

All commands in this stage run in your Ubuntu VM terminal.

Step 1.1: Install Apache

Apache is a web server that will host a simple web page for your lab.

```
sudo apt update
sudo apt install apache2
```

```
vboxuser@Ubuntu:~$ sudo apt install apache2
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Installing:
  apache2

Installing dependencies:
  apache2-bin  apache2-data  apache2-utils  libapr1t64  libaprutil1-dbd-sqlite3  libaprutil1-ldap  libaprutil1t64

Suggested packages:
  apache2-doc  apache2-suexec-pristine  | apache2-suexec-custom

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 23
  Download size: 1,937 kB
  Space needed: 7,565 kB / 90.8 GB available

Continue? [Y/n]
```

Start Apache and enable it to start automatically:

```
sudo systemctl enable --now apache2
```

```
vboxuser@Ubuntu:~$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
vboxuser@Ubuntu:~$
```

Check that it is running:

```
sudo systemctl status apache2
```

```
vboxuser@Ubuntu:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-09-25 19:53:47 UTC; 2min 24s ago
     Invocation: 34d30eb1999e438e8f658fd96949669d
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 10694 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 55 (limit: 9708)
    Memory: 5.6M (peak: 6.3M)
       CPU: 206ms
    CGroup: /system.slice/apache2.service
            └─10694 /usr/sbin/apache2 -k start -DFOREGROUND
              └─10710 /usr/sbin/apache2 -k start -DFOREGROUND
                └─10711 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 25 19:53:47 Ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 25 19:53:47 Ubuntu apachectl[10694]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using
Sep 25 19:53:47 Ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-18/18 (END)
```

Look for **active (running)**. Press **q** to exit the status screen.

Step 1.2: Create a simple test web page

Run this command on Ubuntu:

```
echo '<h1>Ubuntu HTTP Home Lab</h1><p>Testing HTTP traffic from Kali.</p>' | sudo tee
/var/www/html/index.html
```

```
vboxuser@Ubuntu:~$ echo '<h1>Ubuntu HTTP Home Lab</h1><p>Testing HTTP traffic from Kali.</p>' | sudo tee /var/www/html/index.
html
[sudo: authenticate] Password:
<h1>Ubuntu HTTP Home Lab</h1><p>Testing HTTP traffic from Kali.</p>
```

This creates a basic web page. It contains no credentials or sensitive information.

Step 1.3: Find Ubuntu's IP address

```
hostname -I
```

Record the IP address on the VirtualBox network that Kali can reach.

For the example below, I'll use **192.168.64.3**, based on your earlier lab setup. If your Oracle VirtualBox lab uses a different IP, substitute that address.

Check that Apache is listening on port 80:

```
sudo ss -ltnp | grep ':80'
```

```
vboxuser@Ubuntu:~$ sudo ss -ltnp | grep ':80'
LISTEN 0      511          *:80          *:
users:(("apache2",pid=10711,fd=4),("apache2",pid=10710,fd=4),("apac
he2",pid=10694,fd=4))
vboxuser@Ubuntu:~$
```

If UFW is enabled, allow HTTP traffic from Kali's IP address. For example:

```
sudo ufw allow from 192.168.78.5 to any port 80 proto tcp
```

Only use that example if `192.168.78.5` is actually Kali's IP address.

Stage 2 — Start Wireshark on Kali

Step 2.1: Open Wireshark

On Kali, launch Wireshark from the applications menu or run:

```
wireshark
```

Select the network interface connected to the same VirtualBox network as Ubuntu.

If you aren't sure which interface to use, run this in a Kali terminal:

```
ip addr
```

Find the interface with Kali's Ubuntu-reachable IP address.

Step 2.2: Start capturing

Double-click the correct interface to begin capturing.

Once the capture is running, enter this display filter in Wireshark:

```
http
```

This filters for traffic Wireshark recognizes as HTTP. You can also use:

```
tcp.port == 80
```

The second filter is useful if HTTP isn't immediately being decoded.

Stage 3 — Generate HTTP traffic from Kali

Keep Wireshark capturing while you perform these steps.

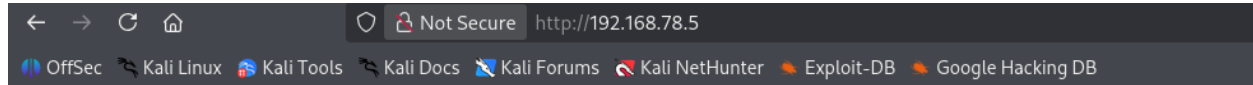
Step 3.1: Access Ubuntu's web page

On Kali, open Firefox and enter this address, replacing the example IP if needed:

```
http://192.168.78.5
```

Make sure you use `http://`, not `https://`.

You should see your Ubuntu HTTP Home Lab page.



Ubuntu HTTP Home Lab

Testing HTTP traffic from Kali.

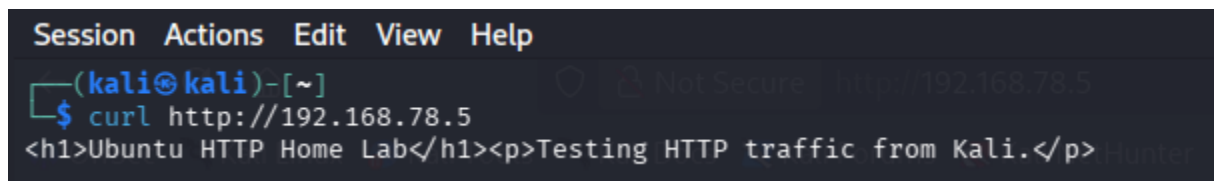
Step 3.2: Generate another request using curl

Alternatively, or in addition to Firefox, run this in the Kali terminal:

```
curl http://192.168.78.5
```

You should see the HTML returned by the Ubuntu web server:

```
<h1>Ubuntu HTTP Home Lab</h1>
<p>Testing HTTP traffic from Kali.</p>
```



This generates a plain HTTP request and response that Wireshark can capture.

Stage 4 — Analyze the HTTP packets

Return to Wireshark and click the red Stop button.

You should now have captured the traffic between Kali and Ubuntu.

Step 4.1: Find the HTTP request

Select a packet identified as an HTTP **GET** request.

In the packet details pane, expand Hypertext Transfer Protocol.

You should see information similar to:

Field	Example
-------	---------

Request method GET

Request URI /

Host 192.168.7
8.5

Protocol HTTP/1.1

No.	Time	Source	Destination	Protocol	Length	Info
543	24.878673877	fd17:625c:f037:2:d7a8:...	2a04:4e42::347	HTTP	386	GET /success.txt?ipv6 HTTP/1.1
545	24.879490367	10.0.2.15	151.101.1.91	HTTP	366	GET /success.txt?ipv4 HTTP/1.1
547	24.902985218	151.101.1.91	10.0.2.15	HTTP	432	HTTP/1.1 200 OK (text/plain)
550	24.902982917	2a04:4e42::347	fd17:625c:f037:2:d7a8:...	HTTP	452	HTTP/1.1 200 OK (text/plain)
2430	64.400880487	192.168.78.3	192.168.78.5	HTTP	400	GET / HTTP/1.1
2432	64.411157693	192.168.78.5	192.168.78.3	HTTP	419	HTTP/1.1 200 OK (text/html)
2434	64.584750423	192.168.78.3	192.168.78.5	HTTP	415	GET /favicon.ico HTTP/1.1
2435	64.596749170	192.168.78.5	192.168.78.3	HTTP	598	HTTP/1.1 404 Not Found (text/html)
2695	236.394011255	192.168.78.3	192.168.78.5	HTTP	144	GET / HTTP/1.1
2697	236.399632658	192.168.78.5	192.168.78.3	HTTP	363	HTTP/1.1 200 OK (text/html)

Step 4.2: Follow the TCP stream

Right-click an HTTP packet and choose Follow → TCP Stream.

You should see the HTTP request sent by Kali and the response returned by Ubuntu. The response may contain your test page's HTML.

This demonstrates that the HTTP content is readable to someone who can capture the network traffic.

```
GET / HTTP/1.1
Host: 192.168.78.5
Authorization: Basic bGFidXNlcjpsYWJwYXNz
User-Agent: curl/8.20.0
Accept: */*

HTTP/1.1 200 OK
Date: Fri, 25 Sep 2026 23:08:14 GMT
Server: Apache/2.4.66 (Ubuntu)
Last-Modified: Fri, 25 Sep 2026 20:00:46 GMT
ETag: "44-65c542c14025e"
Accept-Ranges: bytes
Content-Length: 68
Content-Type: text/html

<h1>Ubuntu HTTP Home Lab</h1><p>Testing HTTP traffic from Kali.</p>
```

Stage 5 — If your assignment requires an HTTP login

The basic web page above demonstrates HTTP traffic, but it does not perform a login. To demonstrate a login, you need a web application or a test page that submits credentials over HTTP.

One simple option is a local HTTP Basic Authentication demonstration in your isolated lab. However, Basic Authentication sends the username and password in a Base64-encoded header, which is not encryption. Wireshark can decode it.

Before setting that up, complete the basic capture first. You can confirm that the HTTP request and response appear in Wireshark.

What to submit for the HTTP capture

- Screenshot of the Ubuntu test page displayed in Kali.
- Screenshot of Wireshark showing the HTTP **GET** request.
- Screenshot of Follow TCP Stream showing the HTTP request and response.
- A short explanation that HTTP sends data without encryption, unlike HTTPS.

Let's get the basic HTTP capture working first. Once you can see the **GET** request in Wireshark, the next step is to add the simulated login and capture the credentials in your lab.

Great! Since you can already see the HTTP **GET** request in Wireshark, your Kali-to-Ubuntu connection and packet capture are working.

Now let's add a simulated HTTP login to your Ubuntu Apache web server. When you log in from Kali, Wireshark will capture the HTTP authentication header so you can examine how credentials are exposed over unencrypted HTTP.

We'll use Apache's built-in HTTP Basic Authentication. This is a simple lab demonstration; it does not require building a full website or installing a database.

What you'll build

Kali Linux

Firefox + Wireshark

Ubuntu

Apache + login

HTTP · TCP port 80 · Lab-only credentials

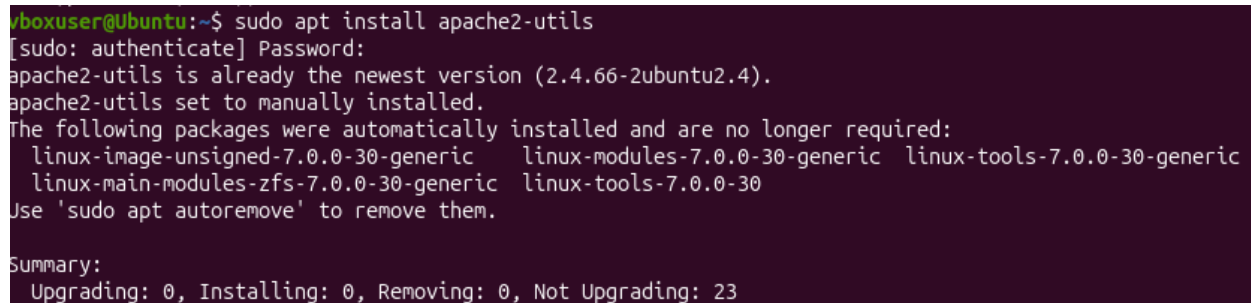
Keep using your existing Ubuntu web server and the same IP address that worked for your [GET](#) request.

Step 1 — Create a test login on Ubuntu

Open the Ubuntu terminal.

Install the Apache password utility:

```
sudo apt install apache2-utils
```

A terminal window with a dark purple background. The prompt is 'vboxuser@Ubuntu:~\$'. The command 'sudo apt install apache2-utils' has been entered. The output shows that the package is already installed and lists several packages that were automatically installed and are no longer required. The summary at the bottom indicates that no packages are being upgraded, installed, or removed, and that 23 packages are not upgrading.

```
vboxuser@Ubuntu:~$ sudo apt install apache2-utils
[sudo: authenticate] Password:
apache2-utils is already the newest version (2.4.66-2ubuntu2.4).
apache2-utils set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 23
```

Create a password file and a test account named [labuser](#):

```
sudo htpasswd -c /etc/apache2/.htpasswd labuser
```

Enter a new, lab-only password when prompted, then confirm it.

For example, you could use a temporary password such as `labpass` — but choose your own and don't reuse a real password.

```
vboxuser@Ubuntu:~$ sudo htpasswd -c /etc/apache2/.htpasswd labuser
New password:
Re-type new password:
Adding password for user labuser
vboxuser@Ubuntu:~$
```

The password is stored in a protected file on Ubuntu. The `-c` option creates the file, so don't use `-c` again when adding more users to that same file.

Step 2 — Configure Apache to require a login

Still on Ubuntu, create a configuration file:

```
sudo nano /etc/apache2/conf-available/lab-login.conf
```

Paste the following configuration:

```
<Directory /var/www/html>
    AuthType Basic
    AuthName "Home Lab Login"
    AuthBasicProvider file
    AuthUserFile /etc/apache2/.htpasswd
    Require valid-user
</Directory>
```

Save and exit Nano:

1. Press `Ctrl + O`.
2. Press Enter.
3. Press `Ctrl + X`.

Enable the configuration:

```

apache2-utils is already the newest version (2.4.66-2ubuntu2.4).
apache2-utils set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 23
vboxuser@Ubuntu:~$ sudo htpasswd -c /etc/apache2/.htpasswd labuser
New password:
Re-type new password:
Adding password for user labuser
vboxuser@Ubuntu:~$ [200~sudo nano /etc/apache2/conf-available/lab-login.conf~
[200~sudo: command not found
vboxuser@Ubuntu:~$ <Directory /var/www/html>
    AuthType Basic
    AuthName "Home Lab Login"
    AuthBasicProvider file
    AuthUserFile /etc/apache2/.htpasswd
    Require valid-user
</Directory>
bash: syntax error near unexpected token `newline'
AuthType: command not found
AuthName: command not found
AuthBasicProvider: command not found
AuthUserFile: command not found
Require: command not found
bash: syntax error near unexpected token `newline'
vboxuser@Ubuntu:~$ which apache2
/usr/sbin/apache2
vboxuser@Ubuntu:~$ ls /etc/apache2/
apache2.conf  conf-enabled  magic  mods-enabled  sites-available
conf-available  envvars  mods-available  ports.conf  sites-enabled
vboxuser@Ubuntu:~$ ls /etc/apache2/conf-available/
charset.conf  localized-error-pages.conf  other-vhosts-access-log.conf  security.conf  serve-cgi-bin.conf
vboxuser@Ubuntu:~$ sudo nano /etc/apache2/conf-available/lab-login.conf
[sudo: authenticate] Password:
vboxuser@Ubuntu:~$

```

sudo a2enconf lab-login

```

vboxuser@Ubuntu:~$ sudo a2enconf lab-login
Enabling conf lab-login.
To activate the new configuration, you need to run:
    systemctl reload apache2
vboxuser@Ubuntu:~$ systemctl reload apache2
vboxuser@Ubuntu:~$

```

Check the Apache configuration for errors:

sudo apache2ctl configtest

You should see:

Syntax OK

```

vboxuser@Ubuntu:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName'
directive globally to suppress this message
Syntax OK

```

```
vboxuser@Ubuntu:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName'
directive globally to suppress this message
Syntax OK
vboxuser@Ubuntu:~$ sudo a2enconf lab-login
Conf lab-login already enabled
vboxuser@Ubuntu:~$
```

Restart Apache to apply the changes:

```
sudo systemctl restart apache2
```

```
vboxuser@Ubuntu:~$ sudo systemctl restart apache2
vboxuser@Ubuntu:~$
```

Your Ubuntu web server should now require authentication before allowing access to the page.

Step 3 — Start a fresh Wireshark capture on Kali

Go back to Kali and open Wireshark.

1. Select the same network interface you used for the earlier HTTP capture.
2. Start capturing.
3. Enter this display filter:

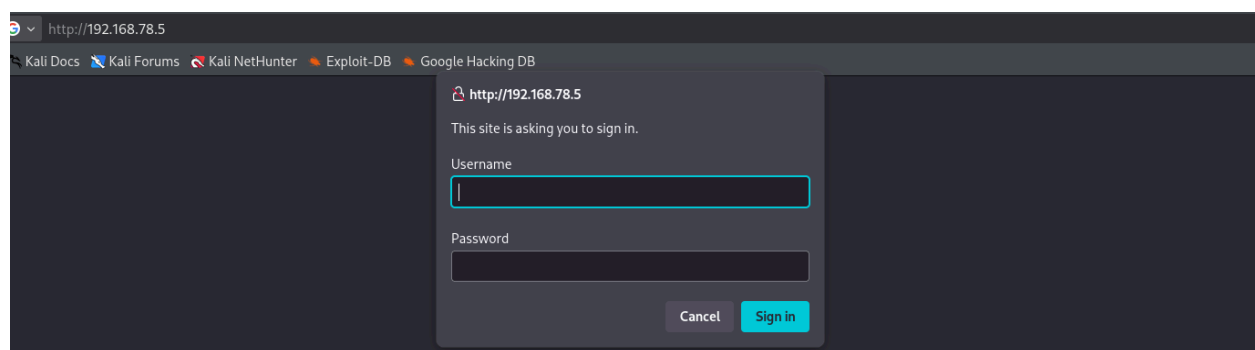
http

Leave Wireshark running.

Step 4 — Log in from Kali

On Kali, open Firefox and navigate to your Ubuntu web server:

<http://192.168.78.5>



You should now see a login dialog.

Home Lab Login

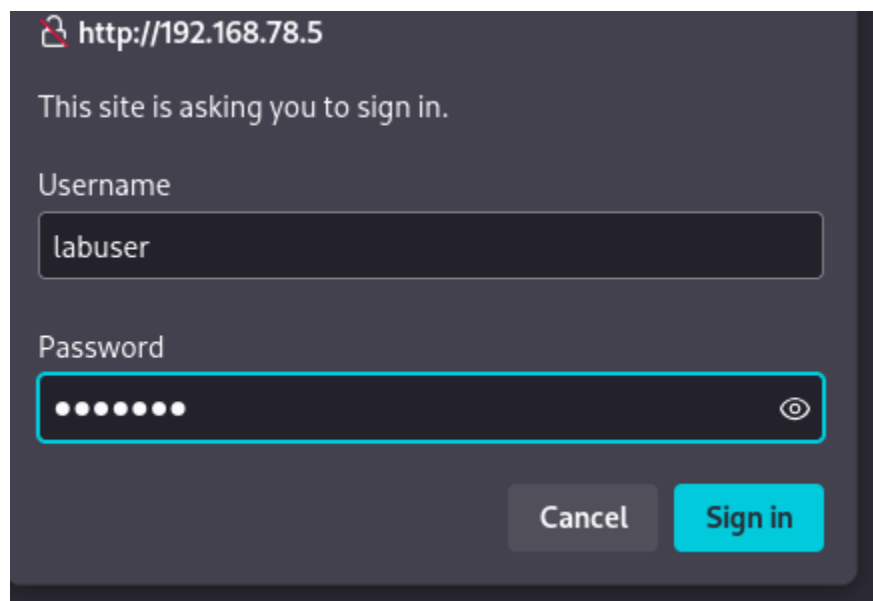
Your browser will display its own authentication dialog.

Username

labuser

Password

Your lab-only password



This is an illustration of the credentials to enter, not a live login form.

Enter **labuser** and the password labpass.

Click Sign in or Log In. You should then see your Ubuntu HTTP test page.

Note: If the browser doesn't prompt you, try a private window or clear the saved credentials for the Ubuntu address or use the terminal on Kali to log in.

```
(kali@kali)-[~]
$ curl -v http://192.168.78.5 -u labuser:labpass
* Trying 192.168.78.5:80...
* Established connection to 192.168.78.5 (192.168.78.5 port 80) from 192.168.78.3 port 48586
* using HTTP/1.x
* Server auth using Basic with user 'labuser'
> GET / HTTP/1.1
> Host: 192.168.78.5
> Authorization: Basic bGFidXNlcjpsYWJwYXNz
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Fri, 25 Sep 2026 23:08:14 GMT
< Server: Apache/2.4.66 (Ubuntu)
< Last-Modified: Fri, 25 Sep 2026 20:00:46 GMT
< ETag: "44-65c542c14025e"
< Accept-Ranges: bytes
< Content-Length: 68
< Content-Type: text/html
<
<h1>Ubuntu HTTP Home Lab</h1><p>Testing HTTP traffic from Kali.</p>
* Connection #0 to host 192.168.78.5:80 left intact
```